

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Open Data Analysis Project

COURSE NAME: Query Processing for Data Science **COURSE CODE:** DSA0503

COURSE OUTCOME COVERED

CO4: Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.

Assessment Tool Weightage: 40%

Total Marks: 20

Code of Conduct

I **Yarra Pawan (192424294)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **Y. Pawan**

Open Data Analysis Project Rubric

Data Collection & Understanding (10 Marks)	Data Cleaning & Preparation (10 Marks)	Exploratory Data Analysis (10 Marks)	Visualization & Dashboard Design (10 Marks)	Demonstration (10 Marks)	Total (50 Marks)

Total Marks Awarded:

Signature of the Course Faculty

Questions:

1. Global COVID-19 Trends and Vaccination Analysis

Using publicly available COVID-19 datasets from sources such as Our World in Data, perform a comprehensive analysis of pandemic trends across different countries. Clean and preprocess the dataset, handle missing values, and explore relationships between infection rates, vaccination coverage, and mortality rates. Create visualizations to identify temporal trends, regional differences, and the effectiveness of vaccination programs. Present findings and recommendations supported by statistical evidence and interactive dashboards.

2. Air Quality and Environmental Impact Assessment

Obtain air quality datasets from World Air Quality Index Project or government open-data portals and analyze pollution levels across cities or regions. Perform exploratory data analysis to identify seasonal variations, pollutant concentration patterns, and correlations between pollutants and meteorological variables. Detect anomalies, visualize geographic distributions using maps, and propose data-driven recommendations for environmental management and public health awareness.

3. Road Accident Analysis and Traffic Safety Evaluation

Using open road accident datasets from Kaggle Datasets or government transportation portals, investigate factors contributing to traffic accidents. Analyze accident frequency with respect to time, weather conditions, road types, and geographic locations. Apply data cleaning techniques, identify high-risk zones, create visual dashboards, and provide recommendations for improving road safety and reducing accident rates based on the insights obtained.

4. Global Education and Literacy Performance Study

Utilize education datasets from UNESCO Institute for Statistics or World Bank Open Data to examine literacy rates, school enrollment, educational expenditure, and academic outcomes across countries. Perform data exploration, correlation analysis, and comparative studies between developed and developing nations. Visualize key indicators using charts and maps, identify trends over time, and provide policy recommendations to improve educational outcomes.

5. Climate Change and Global Temperature Analysis

Using climate datasets from NASA Open Data Portal or NOAA Climate Data Online, conduct an exploratory analysis of global temperature changes and environmental indicators. Investigate long-term trends, seasonal variations, and relationships between temperature anomalies and greenhouse gas emissions. Create informative visualizations, identify significant patterns and outliers, and prepare a detailed report discussing the implications of climate change based on the analyzed data.

Answers

Question No: 1

Question: Using publicly available COVID-19 datasets from sources such as Our World in Data, perform a comprehensive analysis of pandemic trends across different countries. Clean and preprocess the dataset, handle missing values, and explore relationships between infection rates, vaccination coverage, and mortality rates. Create visualizations to identify temporal trends, regional differences, and the effectiveness of vaccination programs. Present findings and recommendations supported by statistical evidence and interactive dashboards.

Dataset Links:

<https://www.kaggle.com/datasets/gpreda/covid-world-vaccination-progress>

<https://www.kaggle.com/datasets/caesarmario/our-world-in-data-covid19-dataset>

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")

vacc = pd.read_csv("country_vaccinations.csv")
owid = pd.read_csv("owid-covid-data.csv")

vacc = vacc.rename(columns={"country": "location"})
vacc["date"] = pd.to_datetime(vacc["date"], errors="coerce")
owid["date"] = pd.to_datetime(owid["date"], errors="coerce")

vacc = vacc.dropna(subset=["date", "location"]).drop_duplicates()
owid = owid.dropna(subset=["date", "location"]).drop_duplicates()

for col in ["total_cases", "total_deaths", "total_vaccinations", "people_fully_vaccinated"]:
    if col in owid.columns:
        owid[col] = owid.groupby("location")[col].ffill()

for col in ["total_vaccinations", "people_vaccinated", "people_fully_vaccinated"]:
    if col in vacc.columns:
        vacc[col] = vacc.groupby("location")[col].ffill()

daily_cols = [c for c in owid.columns if c.startswith("new_")]
owid[daily_cols] = owid[daily_cols].fillna(0)

merged = pd.merge(
    owid,
    vacc[["location", "date", "people_vaccinated_per_hundred",
          "people_fully_vaccinated_per_hundred", "daily_vaccinations"]],
    on=["location", "date"],
    how="left"
)

latest = merged.sort_values("date").groupby("location").tail(1)
top10_cases = latest.nlargest(10, "total_cases")[["location", "total_cases", "total_deaths"]]
print(top10_cases)

if "continent" in merged.columns:
    print(merged.groupby("continent")[["new_cases", "new_deaths"]].sum())
```

```

corr_cols = ["total_cases", "total_deaths", "people_fully_vaccinated_per_hundred",
             "gdp_per_capita", "median_age", "hospital_beds_per_thousand"]
corr_cols = [c for c in corr_cols if c in merged.columns]
corr_matrix = merged[corr_cols].corr()
print(corr_matrix)

latest["case_fatality_rate_%"] = (latest["total_deaths"] / latest["total_cases"]) * 100
print(latest.nlargest(10, "case_fatality_rate_%")[["location", "case_fatality_rate_%"]])

global_trend = merged.groupby("date")[["new_cases", "new_deaths"]].sum()
global_trend.rolling(7).mean().plot(title="Global 7-day Average: New Cases & Deaths")
plt.tight_layout()
plt.show()

plt.figure()
sns.barplot(data=top10_cases, x="total_cases", y="location", palette="Reds_r")
plt.title("Top 10 Countries by Total COVID-19 Cases")
plt.tight_layout()
plt.show()

plt.figure()
sns.heatmap(corr_matrix, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Between Key COVID-19 Indicators")
plt.tight_layout()
plt.show()

if "people_fully_vaccinated_per_hundred" in latest.columns:
    plt.figure()
    sns.scatterplot(data=latest, x="people_fully_vaccinated_per_hundred",
                    y="case_fatality_rate_%",
                    hue="continent" if "continent" in latest.columns else None)
    plt.title("Vaccination Coverage vs. Case Fatality Rate")
    plt.tight_layout()
    plt.show()

```

Output:

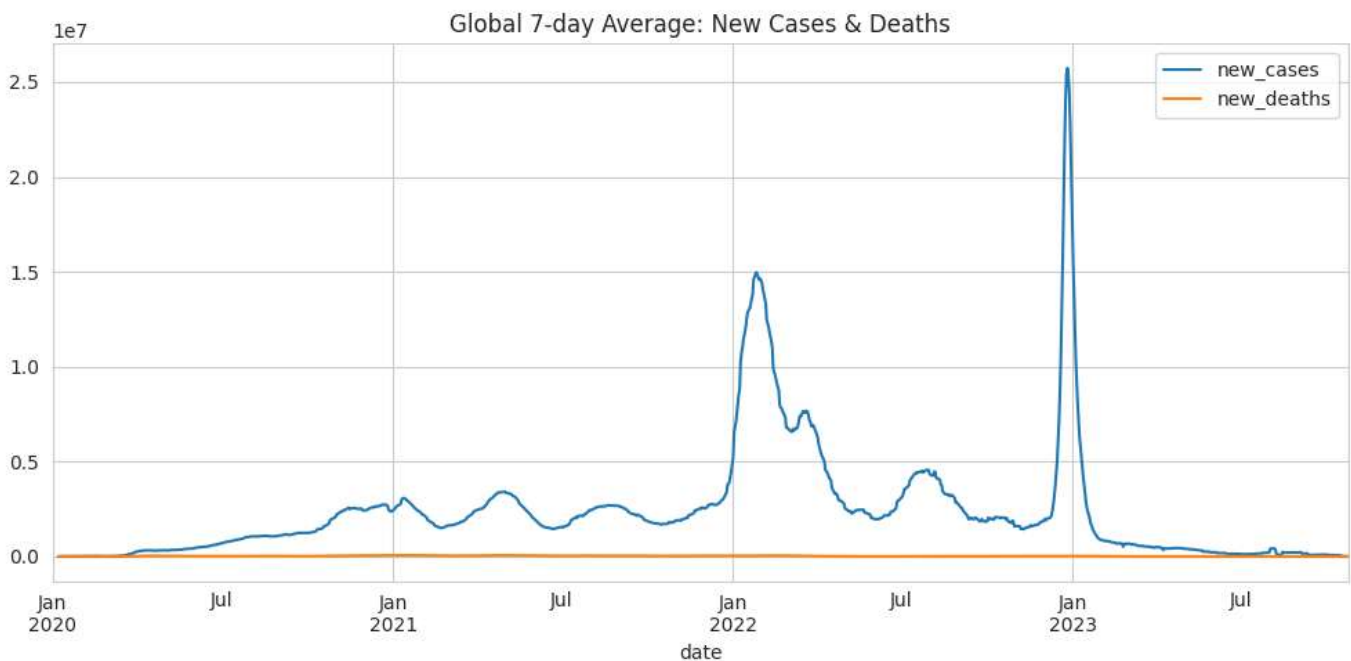
	location	total_cases	total_deaths
345929	World	771407061.0	6972139.0
134308	High income	425988398.0	2914272.0
18019	Asia	300681974.0	1633880.0
98284	Europe	249783754.0	2079783.0
333640	Upper middle income	244463792.0	2666760.0
99674	European Union	184210531.0	1245398.0
225085	North America	124492151.0	1613118.0
330865	United States	103436829.0	1136920.0
62346	China	99315684.0	121742.0
178571	Lower middle income	97423102.0	1339160.0
	new_cases	new_deaths	
continent			
Africa	13115393.0	259034.0	

Asia	300747866.0	1634108.0
Europe	249783755.0	2079783.0
North America	124498040.0	1606145.0
Oceania	14499548.0	29996.0
South America	68843261.0	1359940.0

	total_cases	total_deaths	gdp_per_capita \
total_cases	1.000000	0.938129	0.021657
total_deaths	0.938129	1.000000	0.008642
gdp_per_capita	0.021657	0.008642	1.000000
median_age	0.045863	0.035520	0.648945
hospital_beds_per_thousand	0.017478	-0.003553	0.291031

	median_age	hospital_beds_per_thousand
total_cases	0.045863	0.017478
total_deaths	0.035520	-0.003553
gdp_per_capita	0.648945	0.291031
median_age	1.000000	0.625661
hospital_beds_per_thousand	0.625661	1.000000

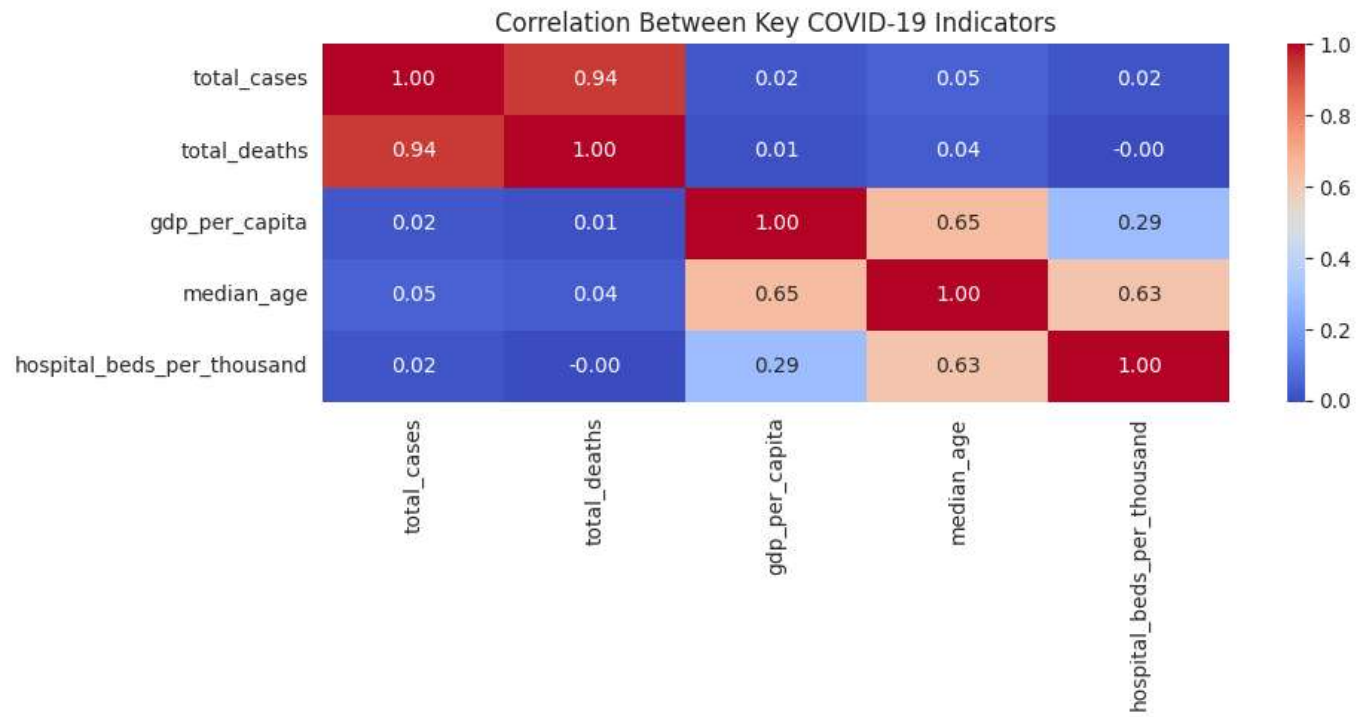
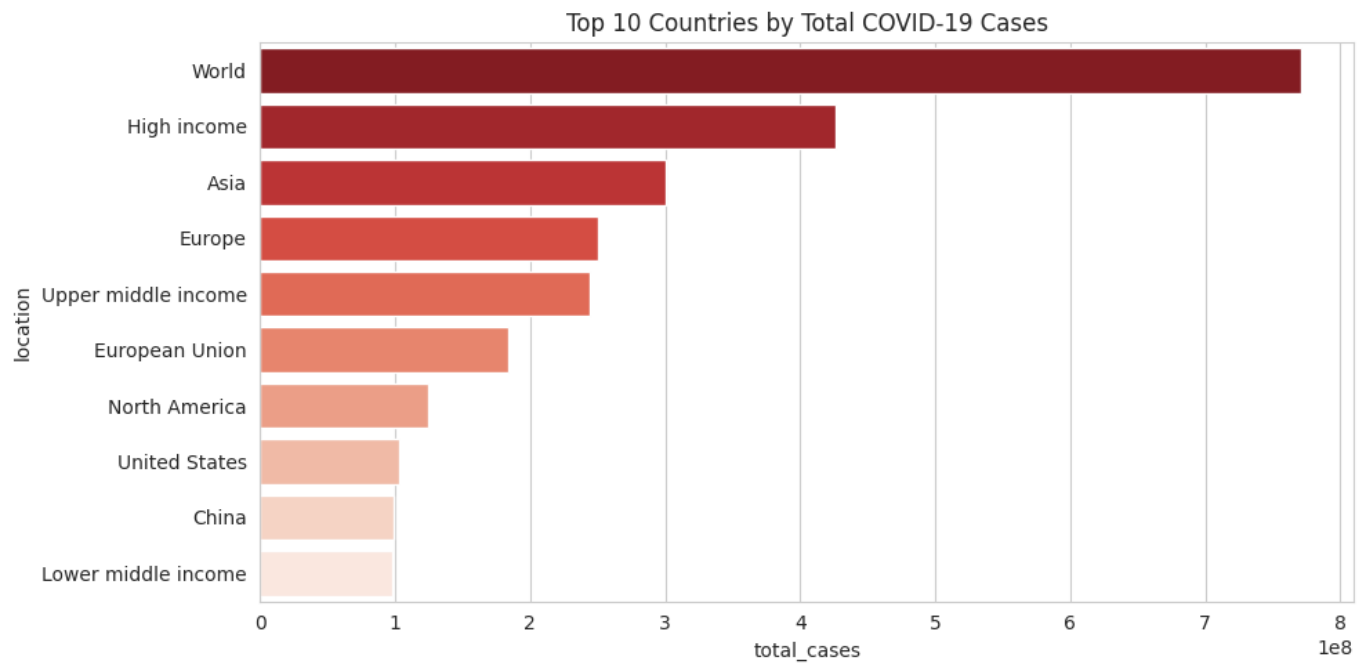
	location	case_fatality_rate_%
347314	Yemen	18.074508
299041	Sudan	7.885237
304587	Syria	5.508246
289341	Somalia	4.979147
245097	Peru	4.904167
87281	Egypt	4.811801
197376	Mexico	4.351036
40181	Bosnia and Herzegovina	4.056765
171636	Liberia	3.707440
1384	Afghanistan	3.505926



/tmp/ipykernel_3760/3689348912.py:58: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(data=top10_cases, x="total_cases", y="location", palette="Reds_r")
```



Question No: 2

Question: Obtain air quality datasets from World Air Quality Index Project or government open-data portals and analyze pollution levels across cities or regions. Perform exploratory data analysis to identify seasonal variations, pollutant concentration patterns, and correlations between pollutants and meteorological variables. Detect anomalies, visualize geographic distributions using maps, and propose data-driven recommendations for environmental management and public health awareness.

Dataset Links:

<https://www.kaggle.com/datasets/hasibalmuzdadid/global-air-pollution-dataset>

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
sns.set_style("whitegrid")
```

```
df = pd.read_csv("global_air_pollution_dataset.csv")
```

```
df = df.dropna(subset=["Country", "AQI Value"]).drop_duplicates()
```

```
top10_polluted = df.nlargest(10, "AQI Value")[["City", "Country", "AQI Value"]]
print(top10_polluted)
```

```
country_avg = df.groupby("Country")["AQI Value"].mean().sort_values(ascending=False)
print(country_avg.head(10))
```

```
numeric_cols = ["AQI Value", "CO AQI Value", "Ozone AQI Value", "NO2 AQI Value", "PM2.5 AQI Value"]
numeric_cols = [c for c in numeric_cols if c in df.columns]
corr_matrix = df[numeric_cols].corr()
print(corr_matrix)
```

```
category_counts = df["AQI Category"].value_counts()
print(category_counts)
```

```
plt.figure()
sns.barplot(data=top10_polluted, x="AQI Value", y="City", palette="Reds_r")
plt.title("Top 10 Most Polluted Cities")
plt.tight_layout()
plt.show()
```

```
plt.figure()
country_avg.head(10).plot(kind="bar")
plt.title("Top 10 Countries by Average AQI")
plt.tight_layout()
plt.show()
```

```
plt.figure()
sns.heatmap(corr_matrix, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Between Pollutant AQI Values")
plt.tight_layout()
plt.show()
```

```
plt.figure()
```

```
category_counts.plot(kind="pie", autopct="%1.1f%%")
plt.title("Distribution of AQI Categories")
plt.ylabel("")
plt.tight_layout()
plt.show()
```

Output:

	City	Country	AQI Value
276	Bahawalnagar	Pakistan	500
470	Rania	India	500
524	Gohana	India	500
611	Gunnaur	India	500
620	Harunabad	Pakistan	500
644	Khetri	India	500
702	Jahangirpur	India	500
714	Kakrala	India	500
720	Kandhla	India	500
830	Mahendragarh	India	500

Country

Republic of Korea	421.000000
Bahrain	188.000000
Mauritania	179.000000
Pakistan	178.788274
United Arab Emirates	163.666667
Aruba	163.000000
Kuwait	162.000000
Qatar	157.500000
India	152.964228
Senegal	152.424242

Name: AQI Value, dtype: float64

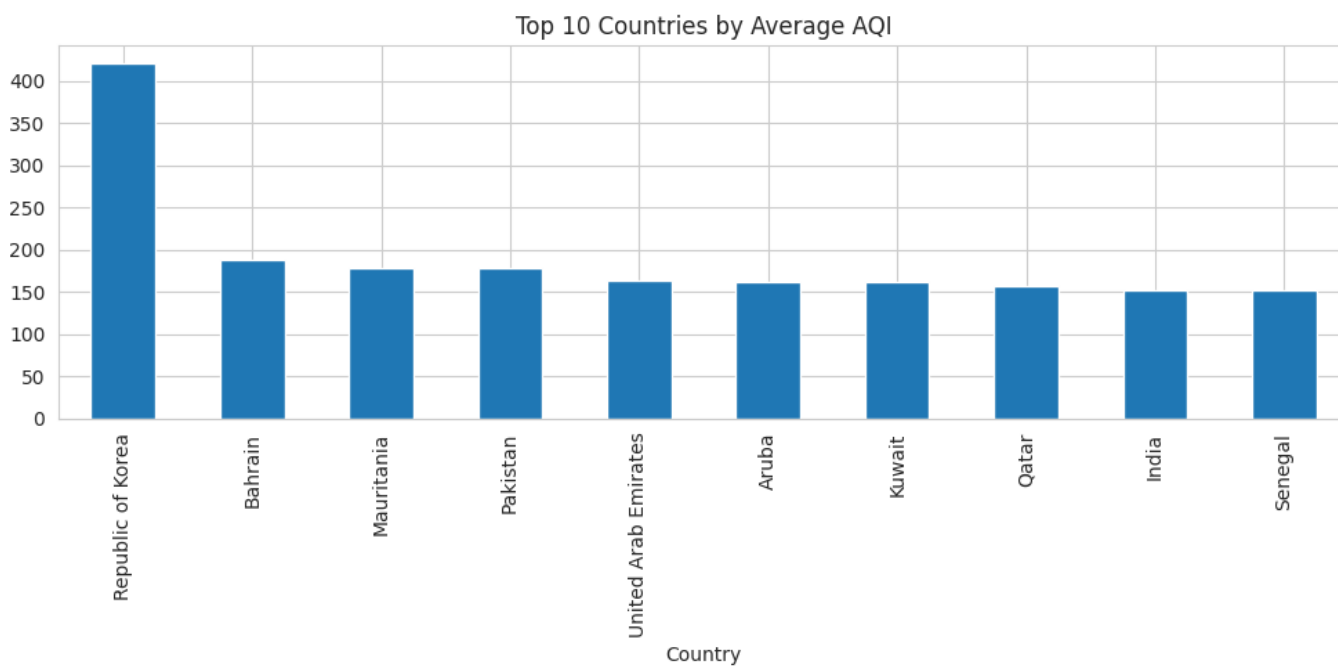
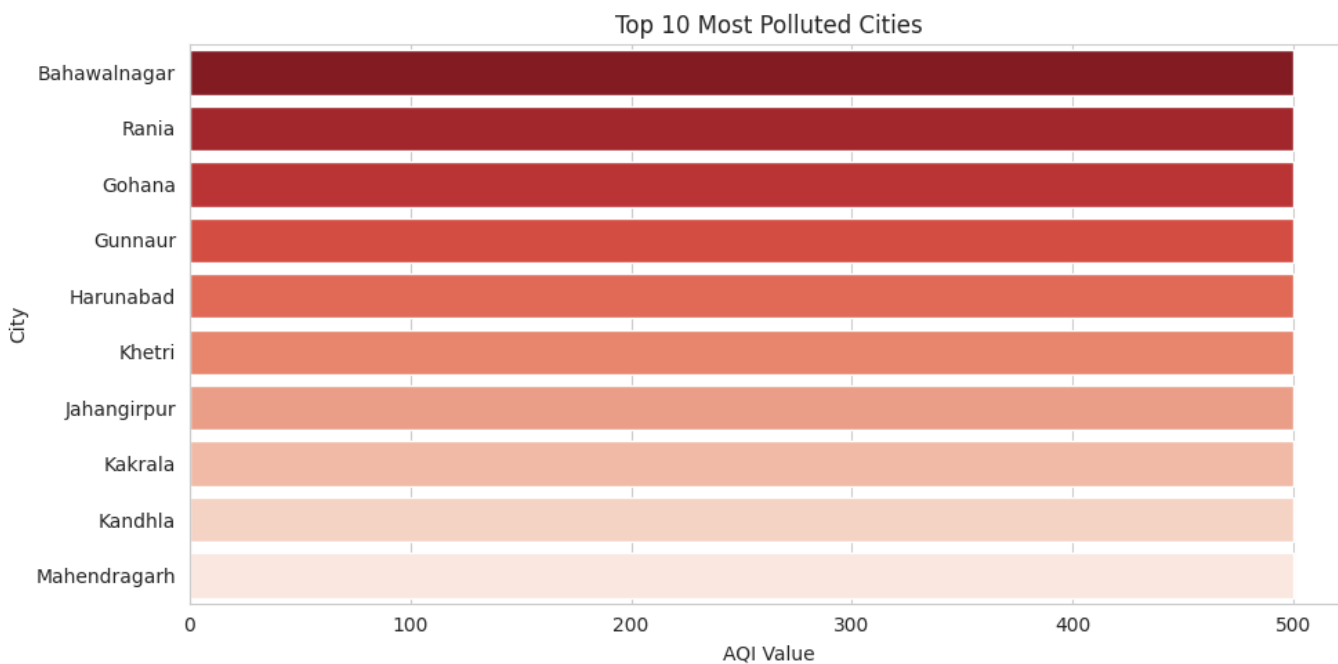
	AQI Value	CO AQI Value	Ozone AQI Value	NO2 AQI Value \
AQI Value	1.000000	0.429643	0.405083	0.230845
CO AQI Value	0.429643	1.000000	0.144833	0.487624
Ozone AQI Value	0.405083	0.144833	1.000000	-0.182930
NO2 AQI Value	0.230845	0.487624	-0.182930	1.000000
PM2.5 AQI Value	0.984518	0.437751	0.340486	0.259084

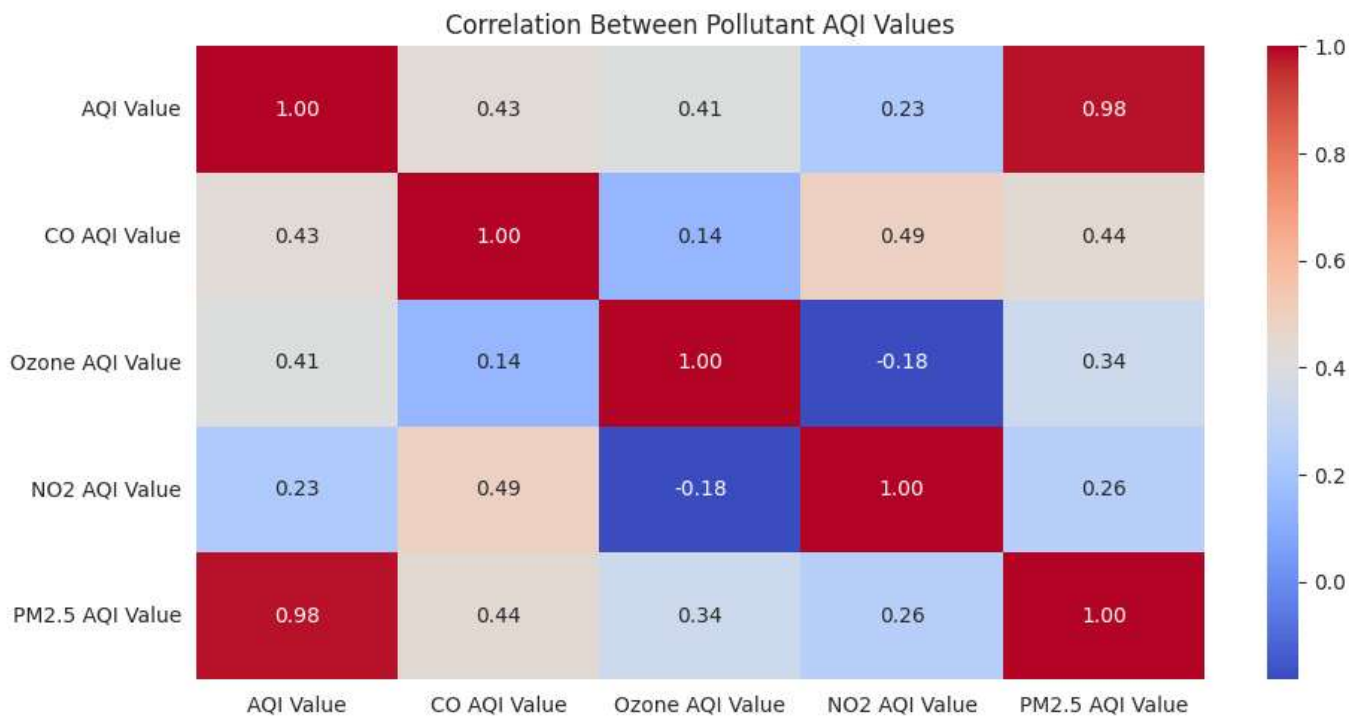
	PM2.5 AQI Value
AQI Value	0.984518
CO AQI Value	0.437751
Ozone AQI Value	0.340486
NO2 AQI Value	0.259084
PM2.5 AQI Value	1.000000
AQI Category	
Good	9688
Moderate	9088
Unhealthy	2215
Unhealthy for Sensitive Groups	1568
Very Unhealthy	286

Hazardous 191
Name: count, dtype: int64
/tmp/ipykernel_3760/1588031744.py:26: FutureWarning:

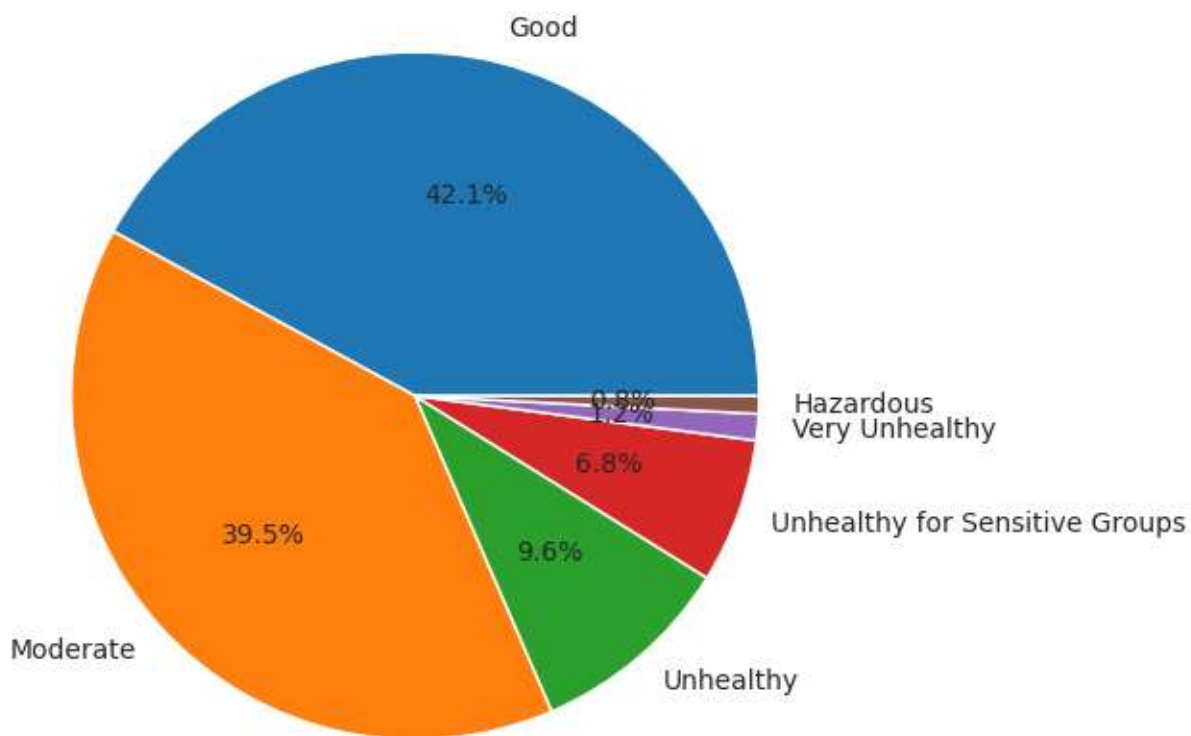
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(data=top10_polluted, x="AQI Value", y="City", palette="Reds_r")
```





Distribution of AQI Categories



Question No: 3

Question: Using open road accident datasets from Kaggle Datasets or government transportation portals, investigate factors contributing to traffic accidents. Analyze accident frequency with respect to time, weather conditions, road types, and geographic locations. Apply data cleaning techniques, identify high-risk zones, create visual dashboards, and provide recommendations for improving road safety and reducing accident rates based on the insights obtained.

Dataset Links:

<https://www.kaggle.com/datasets/saurabhshahane/road-traffic-accidents>

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")

df = pd.read_csv("road_traffic_accidents.csv")

df = df.dropna(subset=["Accident_severity"]).drop_duplicates()

severity_counts = df["Accident_severity"].value_counts()
print(severity_counts)

weather_counts = df.groupby("Weather_conditions")["Accident_severity"].count().sort_values(ascending=False)
print(weather_counts)

road_counts = df.groupby("Road_surface_type")["Accident_severity"].count().sort_values(ascending=False)
print(road_counts)

day_counts = df.groupby("Day_of_week").size()
print(day_counts)

numeric_cols = ["Number_of_vehicles_involved", "Number_of_casualties"]
numeric_cols = [c for c in numeric_cols if c in df.columns]
corr_matrix = df[numeric_cols].corr()
print(corr_matrix)

plt.figure()
severity_counts.plot(kind="bar")
plt.title("Accident Severity Distribution")
plt.tight_layout()
plt.show()

plt.figure()
weather_counts.plot(kind="bar")
plt.title("Accidents by Weather Condition")
plt.tight_layout()
plt.show()

plt.figure()
day_counts.plot(kind="line", marker="o")
plt.title("Accidents by Day of Week")
plt.tight_layout()
plt.show()
```

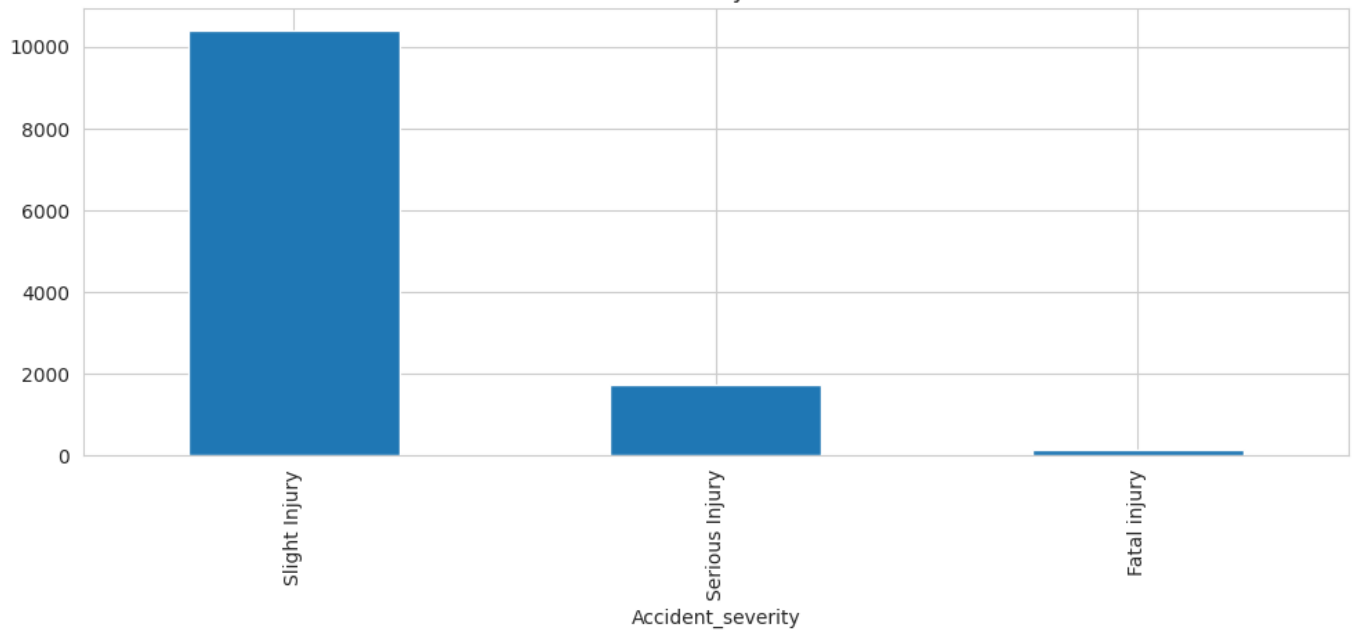
```
plt.figure()
sns.heatmap(corr_matrix, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Between Vehicles Involved and Casualties")
plt.tight_layout()
plt.show()
```

Output:

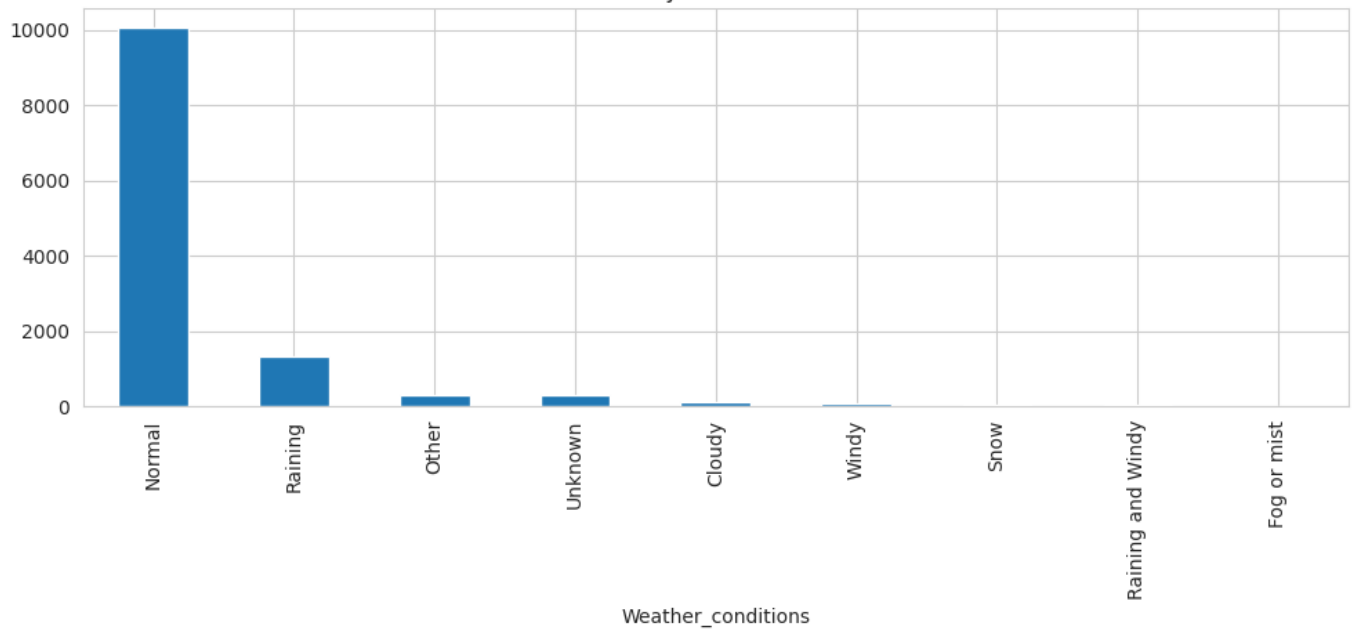
```
Accident_severity
Slight Injury    10415
Serious Injury   1743
Fatal injury     158
Name: count, dtype: int64
Weather_conditions
Normal          10063
Raining         1331
Other           296
Unknown         292
Cloudy          125
Windy           98
Snow            61
Raining and Windy  40
Fog or mist      10
Name: Accident_severity, dtype: int64
Road_surface_type
Asphalt roads    11296
Earth roads      358
Gravel roads     242
Other            167
Asphalt roads with some distress    81
Name: Accident_severity, dtype: int64
Day_of_week
Friday    2041
Monday    1681
Saturday  1666
Sunday    1467
Thursday  1851
Tuesday   1770
Wednesday 1840
dtype: int64

      Number_of_vehicles_involved  Number_of_casualties
Number_of_vehicles_involved      1.000000      0.213427
Number_of_casualties              0.213427      1.000000
```

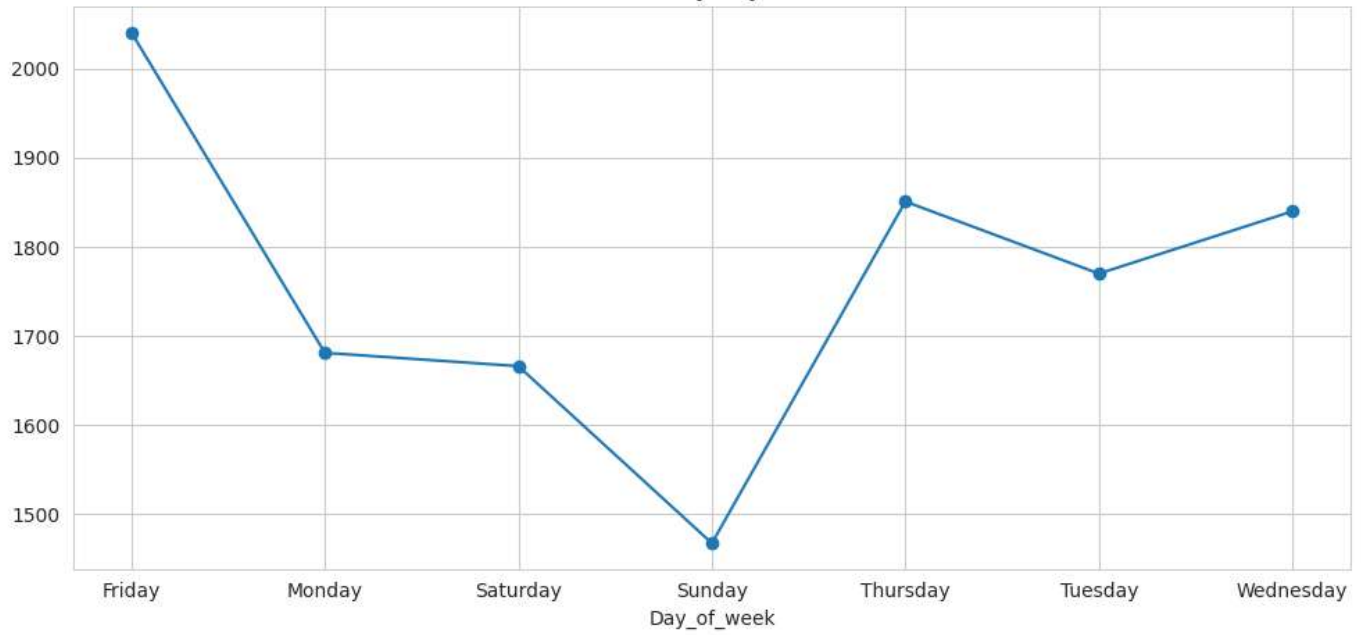
Accident Severity Distribution



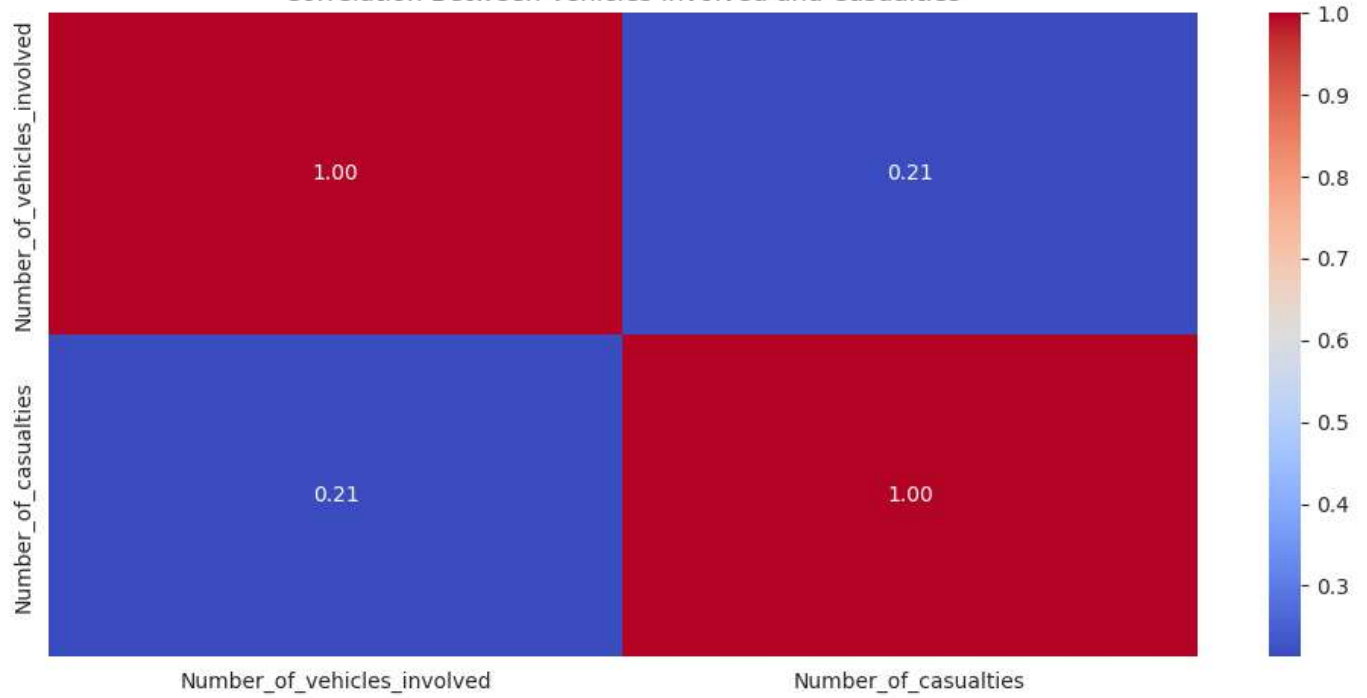
Accidents by Weather Condition



Accidents by Day of Week



Correlation Between Vehicles Involved and Casualties



Question No: 4

Question: Utilize education datasets from UNESCO Institute for Statistics or World Bank Open Data to examine literacy rates, school enrollment, educational expenditure, and academic outcomes across countries. Perform data exploration, correlation analysis, and comparative studies between developed and developing nations. Visualize key indicators using charts and maps, identify trends over time, and provide policy recommendations to improve educational outcomes.

Dataset Links:

<https://www.kaggle.com/datasets/dataanalyst001/world-literacy-rate-by-country>

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")

df = pd.read_csv("world_literacy_rate_by_country.csv")

df = df.dropna(subset=["Country", "Literacy Rate"]).drop_duplicates()

top10 = df.nlargest(10, "Literacy Rate")[["Country", "Literacy Rate"]]
print(top10)

bottom10 = df.nsmallest(10, "Literacy Rate")[["Country", "Literacy Rate"]]
print(bottom10)

if "Continent" in df.columns:
    continent_avg = df.groupby("Continent")["Literacy Rate"].mean().sort_values(ascending=False)
    print(continent_avg)

plt.figure()
sns.barplot(data=top10, x="Literacy Rate", y="Country", palette="Greens_r")
plt.title("Top 10 Countries by Literacy Rate")
plt.tight_layout()
plt.show()

plt.figure()
sns.barplot(data=bottom10, x="Literacy Rate", y="Country", palette="Reds_r")
plt.title("Bottom 10 Countries by Literacy Rate")
plt.tight_layout()
plt.show()

if "Continent" in df.columns:
    plt.figure()
    continent_avg.plot(kind="bar")
    plt.title("Average Literacy Rate by Continent")
    plt.tight_layout()
    plt.show()

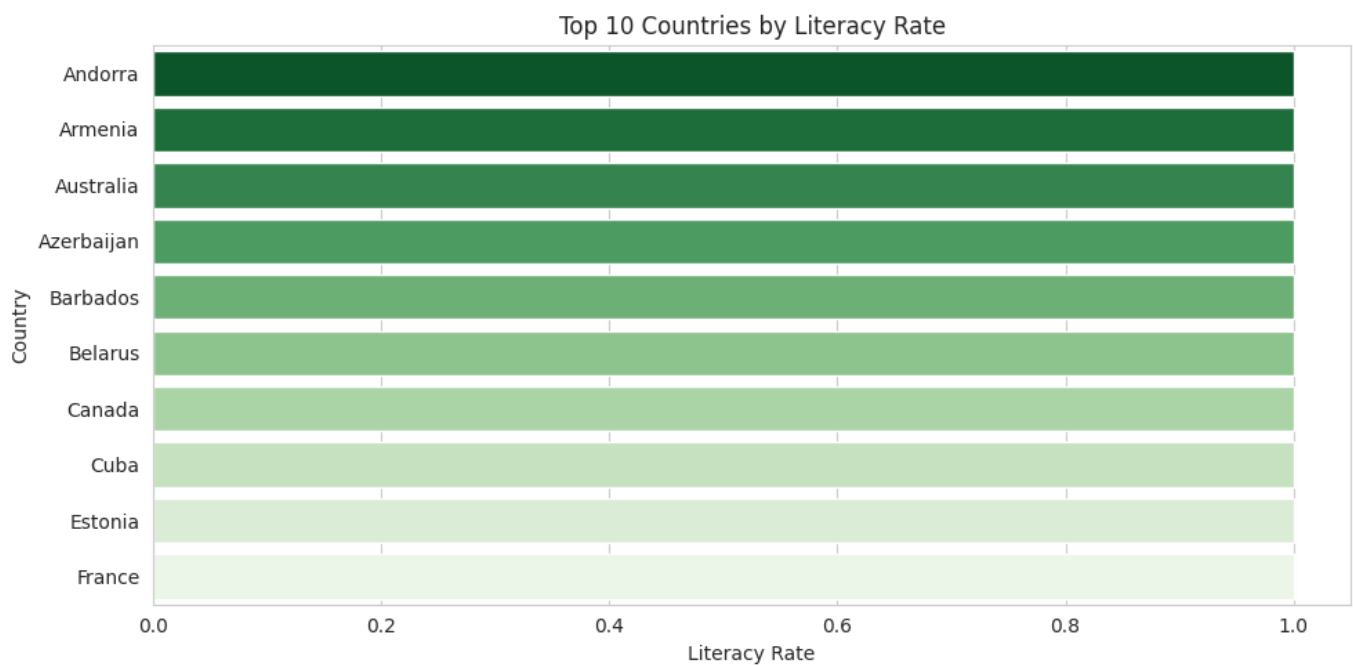
plt.figure()
sns.histplot(df["Literacy Rate"], bins=20, kde=True)
plt.title("Distribution of Literacy Rates Worldwide")
```

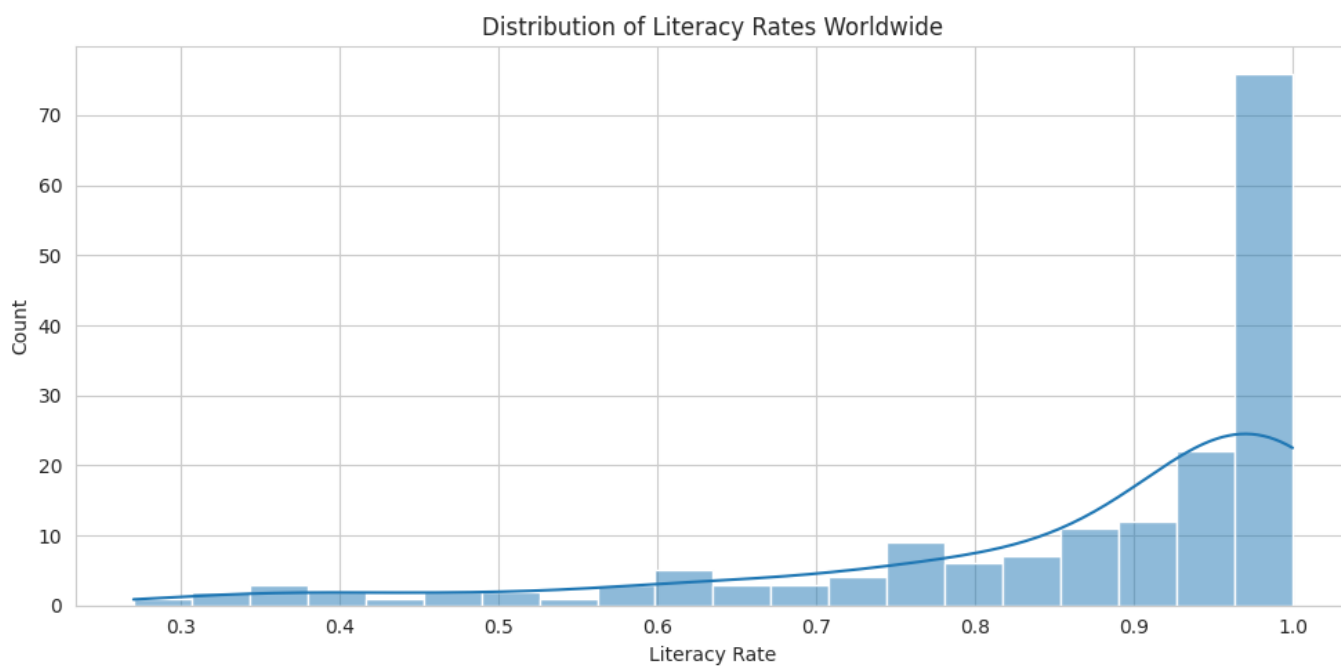
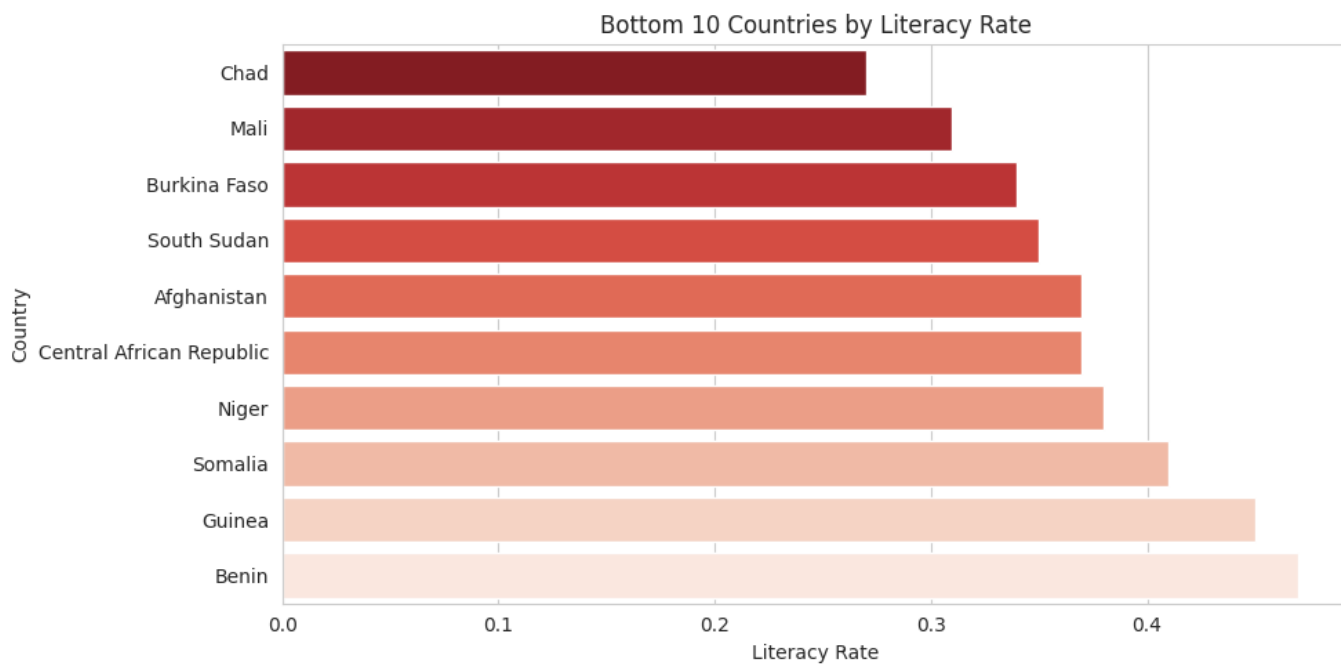
```
plt.tight_layout()
plt.show()
```

Output:

	Country	Literacy Rate
0	Andorra	1.0
1	Armenia	1.0
2	Australia	1.0
3	Azerbaijan	1.0
4	Barbados	1.0
5	Belarus	1.0
6	Canada	1.0
7	Cuba	1.0
8	Estonia	1.0
9	France	1.0

	Country	Literacy Rate
174	Chad	0.27
173	Mali	0.31
172	Burkina Faso	0.34
171	South Sudan	0.35
169	Afghanistan	0.37
170	Central African Republic	0.37
168	Niger	0.38
167	Somalia	0.41
166	Guinea	0.45
165	Benin	0.47





Question No: 5

Question: Using climate datasets from NASA Open Data Portal or NOAA Climate Data Online, conduct an exploratory analysis of global temperature changes and environmental indicators. Investigate long-term trends, seasonal variations, and relationships between temperature anomalies and greenhouse gas emissions. Create informative visualizations, identify significant patterns and outliers, and prepare a detailed report discussing the implications of climate change based on the analyzed data.

Dataset Links:

<https://www.kaggle.com/datasets/bhadramohit/climate-change-dataset>

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")

df = pd.read_csv("/content/climate_change_dataset.csv")

print(df.columns.tolist())

def find_col(keyword):
    for c in df.columns:
        if keyword.lower() in c.lower():
            return c
    return None

year_col = find_col("year")
temp_col = find_col("temp")
co2_col = find_col("co2")
country_col = find_col("country")
sea_col = find_col("sea")
rain_col = find_col("rain")

df = df.dropna(subset=[year_col, temp_col]).drop_duplicates()

yearly_avg_temp = df.groupby(year_col)[temp_col].mean()
print(yearly_avg_temp)

if country_col and co2_col:
    country_avg_co2 = df.groupby(country_col)[co2_col].mean().sort_values(ascending=False)
    print(country_avg_co2.head(10))

numeric_cols = [c for c in [temp_col, co2_col, sea_col, rain_col] if c]
corr_matrix = df[numeric_cols].corr()
print(corr_matrix)

plt.figure()
yearly_avg_temp.plot(kind="line")
plt.title("Average Global Temperature Trend Over Years")
```

```
plt.tight_layout()
plt.show()
```

```
if country_col and co2_col:
    plt.figure()
    country_avg_co2.head(10).plot(kind="bar")
    plt.title("Top 10 Countries by Average CO2 Emissions")
    plt.tight_layout()
    plt.show()
```

```
plt.figure()
sns.heatmap(corr_matrix, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Between Climate Indicators")
plt.tight_layout()
plt.show()
```

```
if co2_col:
    plt.figure()
    sns.scatterplot(data=df, x=co2_col, y=temp_col)
    plt.title("CO2 Emissions vs Temperature")
    plt.tight_layout()
    plt.show()
```

Output:

['Year', 'Country', 'Avg Temperature (°C)', 'CO2 Emissions (Tons/Capita)', 'Sea Level Rise (mm)', 'Rainfall (mm)', 'Population', 'Renewable Energy (%)', 'Extreme Weather Events', 'Forest Area (%)']

Year

2000	20.501887
2001	20.117073
2002	21.433333
2003	18.219512
2004	18.804000
2005	19.532432
2006	19.805128
2007	20.562500
2008	19.144444
2009	19.730303
2010	17.963333
2011	18.856818
2012	17.569048
2013	18.829032
2014	20.834091
2015	19.916279
2016	21.160784
2017	19.478125
2018	23.250000
2019	20.402500
2020	19.044186

2021 20.076087

2022 18.960000

2023 21.107143

Name: Avg Temperature (°C), dtype: float64

Country

UK 12.650769

Indonesia 11.021333

France 10.966667

Japan 10.960317

USA 10.882192

Brazil 10.838806

China 10.835821

India 10.694286

Argentina 10.447761

Canada 10.113433

Name: CO2 Emissions (Tons/Capita), dtype: float64

Avg Temperature (°C) \

Avg Temperature (°C) 1.000000

CO2 Emissions (Tons/Capita) 0.012323

Sea Level Rise (mm) 0.059005

Rainfall (mm) -0.004532

CO2 Emissions (Tons/Capita) Sea Level Rise (mm) \

Avg Temperature (°C) 0.012323 0.059005

CO2 Emissions (Tons/Capita) 1.000000 -0.038815

Sea Level Rise (mm) -0.038815 1.000000

Rainfall (mm) 0.013306 0.022104

Rainfall (mm)

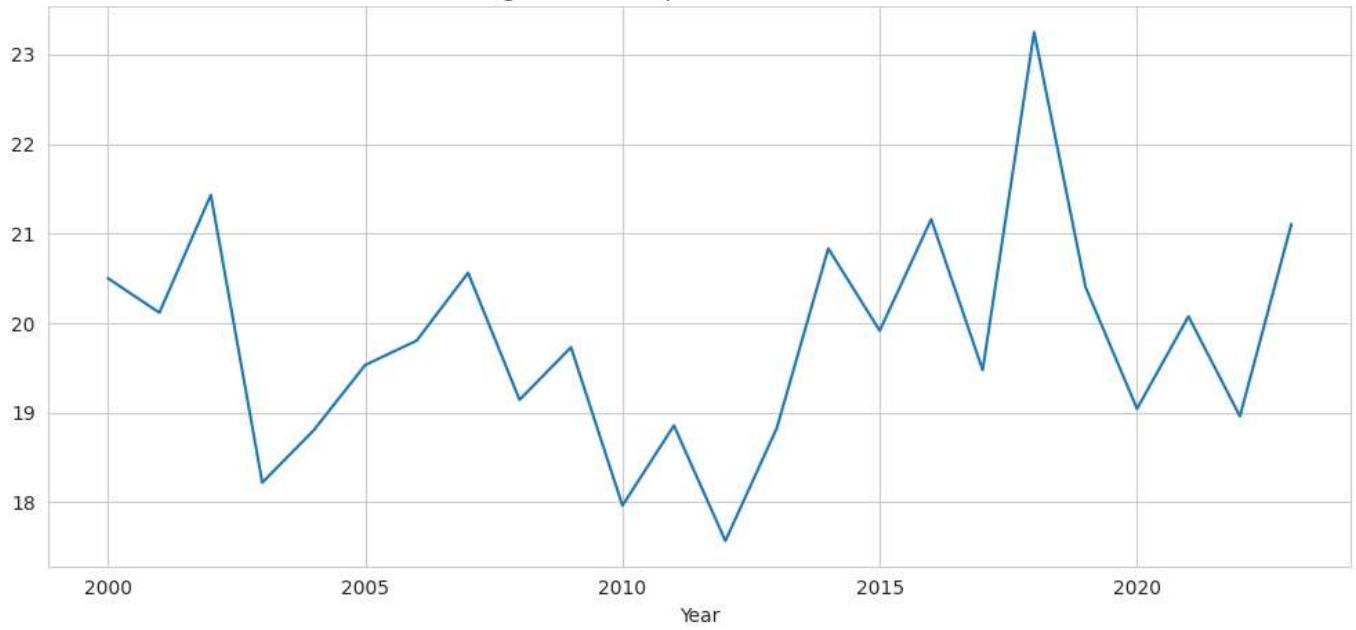
Avg Temperature (°C) -0.004532

CO2 Emissions (Tons/Capita) 0.013306

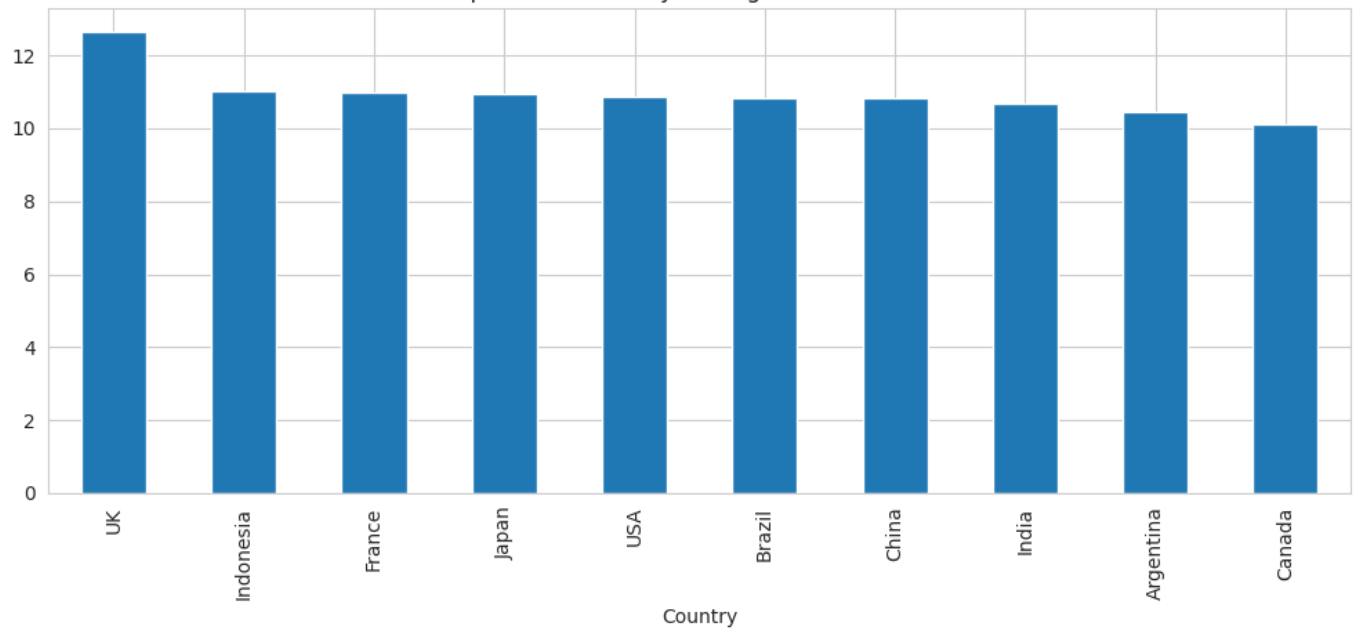
Sea Level Rise (mm) 0.022104

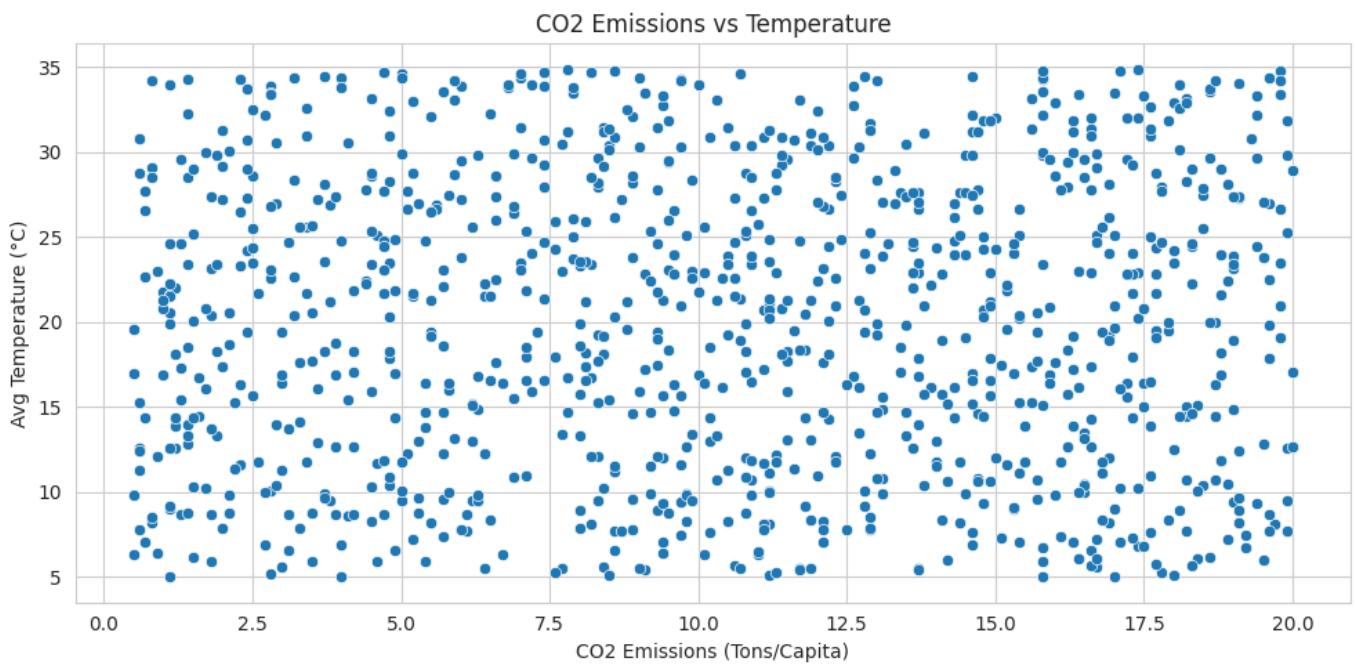
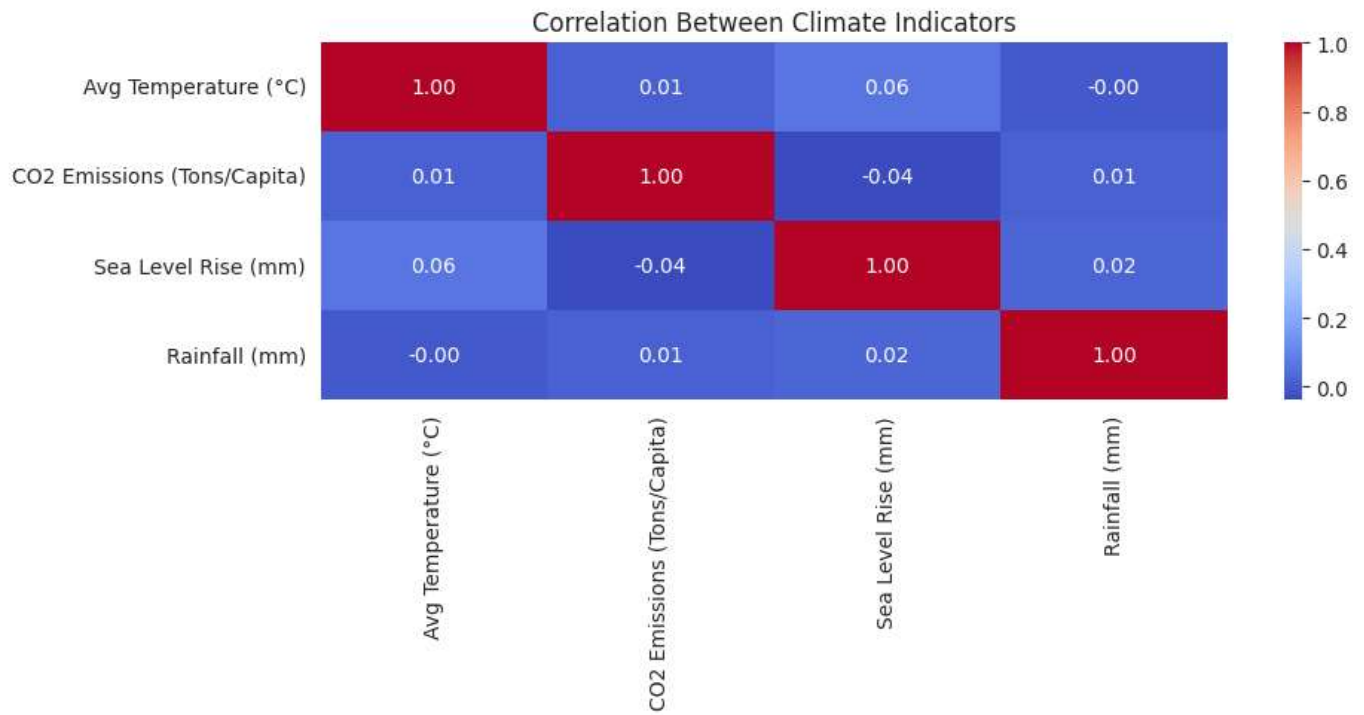
Rainfall (mm) 1.000000

Average Global Temperature Trend Over Years



Top 10 Countries by Average CO2 Emissions





https://colab.research.google.com/drive/18kDGRMSSbV4piYebfRM4vA3w5g_K4jDy?usp=sharing